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Skin-Integrated Wireless Odor Message Delivery Electronics for the Deafblind

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Abstract

Deaf-blindness limits daily human activities, especially, interactive modes of audio and visual perception. Although the developed standards have been verified as alternative communication methods, they are uncommon to the non-disabled due to the complicated learning process and inefficiency in terms of communicating distance and throughput. Therefore, the development of communication techniques employing innate sensory abilities including olfaction related to the cerebral limbic system processing emotions, memories, and recognition has been suggested for reducing the training level and increasing communication efficiency. Here, a skin-integrated and wireless olfactory interface system exploiting arrays of miniaturized odor generators (OGs) based on by melting/solidifying odorous wax to release smell, is introduced for establishing an advanced communication system between deafblind and non-deafblind. By optimizing the structure design of the OGs, each OG device is as small as 0.24 cm^3 (length $\times$ width $\times$ height of $11 \text{ mm} \times 10 \text{ mm} \times 2.2 \text{ mm}$), enabling to integrate up to 8 OGs on the epidermis between nose and lip for direct and rapid olfactory drive with weight of only 24.56 g. By generating single or mixed odors, different linked messages could be delivered to a user within a short period in a wireless and programable way. By adopting the olfactory interface message delivery system, the recognition rates for the messages have been improved 1.5 times that of the touch-based method, while the response times were immensely decreased 4 times. Thus, the presented wearable olfactory interface system exhibits great potential as an alternative message delivery method for the deafblind.

Keywords: *flexible electronics, wearable electronics, smell generation, wireless message delivery, human machine interfaces*

Introduction

Deafness and blindness are significant physical disabilities that extensively limit the daily life activities by tremendously narrowing the interacting methods including audio and visual perception.¹⁻³ Thus, patients, suffering from complete or partial vision and hearing loss, may encounter uncomfortable situations especially when they have to communicate with others who are not trained in augmentative and alternative communication including Braille or sign language.^{4,5} Clinically, a number of approaches have been developed based on tactile methods (e.g., finger Braille and Tadoma) for facilitating deafblind patients to establish partial communication with others.⁶⁻¹⁰ However, up to now, there are still unsettled obstacles including the complicated learning process, low communication efficiency, short communication distance induced by tactile inputs, and low throughput (one-to-multi crosstalk).¹¹⁻¹³ For example, for sign language, although repeated training was conducted on patients, they showed difficulties in reproducing and embracing the specific gestures, while it took 18 months to fluently manage the sign language.^{14,15} In addition, for Braille reading, the learning rate of this tactile communicating method was highly dependent on physical conditions such as age, gender, handedness, and the presence of haptic and visual aids.^{16,17} Furthermore, for people who were beginner to Braille reading, mistakes in reading still existed even after a few months of training.¹⁸ Meanwhile, the conventional alternatives are still limited in low communication efficiency and throughput. To improve communication efficiency while reducing burdens on training or other cognitive efforts, techniques for enhancing remaining sensory abilities, such as taste and olfaction, have recently emerged as additional strategies for supporting deafblind patients.¹⁹⁻²¹

Olfaction, as an innate sensation, plays a vital role in human interaction with the surrounding environment associating with the cerebral limbic system processing emotions, recognition, and memories.²²⁻²⁶ As odor recognition can be achieved with relatively minor cognitive efforts and conscious attempts through olfactory sense, external information is able to be obtained efficiently without the aid of visual and audio perception.^{24,27} In addition, it is reported that olfaction is possible to be used as a complementary module for providing additional information (olfactory cues) to enhance learning, memory, and comprehension in education

and life for deafblind individuals, such as identifying people, spaces, objects, and activities.²⁸⁻³² Compared to the conventional alternatives with a great command of languages in terms of delivering complex and customized messages,^{6, 8, 10} as the olfaction interfaces are in the early stage, they are limited by the number of smells that can be accurately discriminated.^{22, 33} Accordingly, recently, a number of research has been conducted on olfactory technologies for enhancing performance and increasing the number of firmly detectable odors.^{34, 35} Furthermore, the olfaction interface provides unidirectional and regional propagation, which enables noncontact and wireless communication and one-to-multiple crosstalk, unlike the aforementioned alternatives. Despite their high potentiality, up to date, olfaction systems that most modalities are built on rigid and bulky platforms are still with critical flaws, hindering them to be widely applied for aiding deafblind patients, including (1) inaccurate control of odor generation that harshly decreases the recognition rate,³⁶ and (2) bulky in size and rigidity restraining them from long-term wearability and real-time wireless message delivery.^{37, 38} Therefore, to overcome the mechanical mismatches and increase the accuracy of the control and recognition rate, the flexible and wearable form factor, which is with mechanical and physical characteristics including light-weight, bendability, twistability, stretchability, and ultra-thin thickness, for the olfaction interfaces has been suggested for enabling long-term wearable application with a prompt responding to external input.³⁹⁻⁴⁵

Herein, we report a skin-integrated and wireless olfactory interface system based on arrays of miniaturized odor generators (OGs) for building up a universal communication method, olfaction, between the deafblind and non-deafblind subjects. By linking odors to specific messages, the wearable olfactory electronics could realize a wireless odor-based information delivery to the deafblind who generally obtains information by touch during their daily life. As the odor generation unit of the whole system, OGs could accurately release or shut down various odors by melting or solidifying odorous wax in a short time (~ 0.6 s), in which the printed electrode serves as heaters and the mechanical actuators acts as the active cooling system. Compared to the OGs reported by our group very recently,⁴⁶ the OGs reported here adopt a simplified structure to realize a smaller overall size (length $\times$ width $\times$ height of 11 mm $\times$ 10 mm $\times$ 2.2 mm), where the heating electrode also serves as a temperature sensor by continuously monitoring its resistance variation while the previously reported OGs

managed thermistor as thermal sensor and heater separately. Following, the fabricating process was streamlined; meanwhile, the size and weight of the OG unit were reduced. Due to the advancement in the structure design, which improved the electromagnetic cooling system, a faster cooling effect could be established to rapidly turn off the release of odor on purpose, and as a result, the response time of the OG in increasing and dropping down temperature could reach up to 0.6 s and 1.36 s respectively. In addition, a 3D-printed shell is mounted onto the heating electrode of OGs for preventing the melted wax from flowing outside during practical applications while ensuring the continuous and safe operation of OGs. As a result, the miniaturized design, lightweight (0.4 g), advanced electrical performance, and robust structure of OGs contribute to the optimal olfaction system for achieving a high OG array density ($0.72/\text{cm}^3$) and stable operation (~ 10000 working cycles) in a long-term practical application where the system could be directly mounted onto skin surface between the human nose and upper lip. Moreover, by training 31 volunteers' olfaction ability through the olfactory interface, the average odor messages recognition rate and response time had been obviously enhanced compared to those by the traditional handwriting method, and therefore, its great potentiality as an alternative message delivery method for the deafblind was proved simultaneously.

Results

As shown in **Figure 1A**, the skin-integrated wireless odor messages delivery electronics system adopts a multilayer stacking structure. Firstly, the core parts of the olfaction interface are constructed with an array of 8 odor generators (OGs) that synchronizes or desynchronizes the modulation of multiple odors with spatiotemporal precision. Then, a flexible printed circuit boards (FPCBs) (**Figure S1**) is installed with surface-mounted device (SMD) types of electrical components and encapsulated by two polydimethylsiloxane (PDMS) layers (0.4 mm, ~ 145 kPa) at the top and bottom for protecting the inner circuit from unpredictable external stimuli such as bending and twisting. The OG unit with the overall size of $11\text{ mm} \times 10\text{ mm} \times 2.2\text{ mm}$ (length $\times$ width $\times$ height) employs a heating platform filled with odor/wax mixtures and a thermal actuator for establishing a continuous, rapid, and graded odorant release and an electromagnetic active cooling system to swiftly extinguish odor release by cooling down the OG unit, respectively (**Figure S2**). Due to the application of the

odor/wax mixtures, the lifespan of the OGs was further extended as when the smell is exhausted or required to be replaced, the melted wax can be removed by exploiting cotton, and then, additional wax with odor can be simply refilled without replacing or damaging the OG unit. **Figure 1B** presents the structure of the OG unit in the multiple layers (**Figure S3**). From the top to bottom, a biocompatible paraffin wax layer containing various odor components to generate smell is covered by a rigid 3D printed shell structure for protecting the melted wax from leakage during motion and daily activities (**Figure S4**). Next, at the bottom layer of the polyimide (PI, 2 μm thick), a metallic electrode composed of gold (Au) (200 nm) and chromium (Cr) (40 nm) that acts as a controllable heating source to melt odorous paraffin wax for releasing the smell and simultaneously serves as a temperature sensor for precisely monitoring the heating temperature of the electrodes is mounted on the top layer of polyethylene terephthalate (PET) film (25 μm) (**Figure 1C**).

Furthermore, at the bottom of the cantilever-shaped PET layer, a copper (Cu) coil (480 turns; wire diameter of 50 μm , length of 0.8 mm) is fixed for providing functionality as a mechanical actuator while the PET provides a supporting layer for both thermal and mechanical actuators. The PET platform is attached to the rectangular hollow PDMS ring with 1 mm thickness wrapping a permanent magnet disc (5 mm $\times$ 5 mm $\times$ 1 mm) affixed on the most bottom PET substrate (25 μm). The integrated electromagnetic active cooling system is designated for promoting rapidly heating up and cooling down of the heater while the heating platform is lifted up in the air and down to be in contact with the magnet, respectively. According to the direction of electrical current through the Cu coil, the electromagnetic field could be programmed to change the direction of the applied force applied to the heating platform. To prove the mechanical stability of the PET film and heating electrode, they were bent (90°) and twisted (53.8°) (**Figure S5**). As a result, the maximum stresses of PET and metallic heating electrode are both within the corresponding yield strength limits.

At last, the whole system is operated by two symmetric electrical circuits installed with a number of different electrical components including microcontroller units (MCU, Atmega328P) for controlling the system, two different types of N-channel metal-oxide semiconductors (NMOS) and P-channel metal-oxide semiconductors (PMOS) to build separate high power and low power load switches for regulating the

temperature of the heaters, light-emitting diodes (LED) for avoiding electrical power interruption between low and high power, 24-bit analog-to-digital converter (ADC, ADS1220) to increase the resolution of the thermal sensing, decoders for controlling the direction of the current flow through the Cu coils, a Bluetooth module (WH-BLE106) for wireless communication to an external device, and four different low-dropout (LDO) regulators with the output voltages of 2.6 V, 3.3 V, 5 V, and 5.6 V for the low power line, Bluetooth module, MCU, and high power line respectively (**Figure 1G**). For establishing the temperature management system of the olfaction interface, the closed-loop system with 200 ms of systemic delay is programmed on the MCU (**Figure 1D**). When the target temperature is selected, the system automatically compares the currently and previously selected temperatures and if the actual temperature is lower than the target, the high-power line that is with sufficient heating power to heat the heater rapidly is turned on until the actual temperature reaches the target. If the actual temperature is higher than the target, the low power line which the temperature of the heater decreases rapidly on account of insufficient heating power is turned on while the cooling system is activated by switching the direction of the current flow into the Cu coil to help the dissipation of heat and then, when it reaches the target, the cooling system is turned off (**Figure S6**). As soon as the thermal management system achieves the target temperature, the system enters the temperature-stabilizing state by shifting the power lines accordingly. Due to the presence of two different power lines, the temperature can be calculated back constantly regardless of the heating and cooling phases. Subsequently, the working principle of the smell-generating procedures of the OG is demonstrated with aid of a schematic illustration (**Figure 1E**). During the silent state, the heating platform is kept hanging in the air with the heating temperature stabilized at around 45°C. Here, to realize a fast response time in release odor, 45 °C had been chosen for the stand-by heating temperature as it is highly close to the melting point of the wax which is around 50 °C. As a consequence of a relatively longer systemic delay compared to our previously reported OGs,⁴⁶ more significant thermal fluctuations during stabilizing state resulted due to the extended high and low-power switching time.

To generate odor, a constant heating power is supplied to the heating electrode to immediately raise the temperature of the odorous wax to its melting point and then, the molecules of the wax are diffused into the air

for releasing a specific smell. Conversely, to terminate the odor generation, the cantilever is bent down toward the magnet to induce contact between the heating platform and the magnet for dissipating heat swiftly and therefore, the melted wax is solidified back. Furthermore, the precise temperature-controlling capability of the olfaction system for establishing the controlled spatial and temporal distribution of heat and high stability for the magnitude of the smell generating is verified by monitoring the thermal distribution of the OG unit with different target heating temperatures. As shown in **Figure 1F**, the measured temperatures of the OG unit are highly close to the corresponding preset target temperatures (45 °C, 50 °C, 55 °C, and 60°C) and this indicates the stable and accurate operation and highly controllable thermal management of the thermal actuator during the practical application (**Figure S7**). Thereafter, the representative prototypes of the olfaction interface with FPCBs and eight OGs arrays installed with the prementioned electrical components with and without PDMS encapsulation respectively were assembled (**Figure 1G and H**). Then, the encapsulated device was attached to the human skin between the nose and upper lip to examine its mechanical stability against the mouth motion (**Figure S8**). While the mouth was moving, the device was successfully affixed to the position without any distortion or failure. Furthermore, for testing the adhesive property of the device, the device was applied to volunteers with and without double-sided tape at the bottom of the device to enhance the adhesion while the volunteers were executing specific physical activities, including running, walking, and sitting (**Figure S9**). As shown in **Figure S9A and B**, without the aid of double-sided tape, intensely low adhesion times were measured under running and walking for both male and female volunteers (male: 16.5 min and 22.3 min, and female: 14.5 min and 21.5 min, respectively). However, sitting that is limited in body movement and with double-sided tape, the device could attach to the human epidermis for more than 3 hr which is the optimum operation duration for the system. Thus, by adopting double-sided tape, the adhesive property of the device was immensely improved, while it enabled to cover the strenuous exercise during application. As a result, the wearability and mechanical strength against the user's mouth activity had been approved.

Next, for investigating the operational performance of the olfaction interface and the optimal electrical power for the heating source and the structural dimension for the heating platform, a series of experiments and

characterizations have been conducted under controlled parameters. To investigate the temperature-sensing characteristics of the OG, the electrical power matched with the high-power line was applied to the heating electrode of the OG at room temperature while the temperature of the heater had been monitored by a commercial infrared thermometer (MLX90614). As a result, a considerably linear relationship between the resistance of the electrode (from 78.47 to 95.11 Ω) and the heating temperature (from 22.11 to 71.89 $^{\circ}\text{C}$) was exhibited (**Figure 2A**). Therefore, a fitted curve line could be extracted from the graph to formulate an equation for calculating the temperature at the specific resistance.

Subsequently, the performance of the electromagnetic active cooling system was analyzed by comparing the heating and cooling response times of the OG with and without the cooling system. The heater was heated up from the stand-by heating temperature (45 $^{\circ}\text{C}$) to the target temperatures (50 $^{\circ}\text{C}$, 55 $^{\circ}\text{C}$, and 60 $^{\circ}\text{C}$ respectively), then cooled down to the stand-by heating temperature after reaching the corresponding temperatures to monitor the thermal management of the heater at different target thermal ranges, where the powers into the heater and electromagnetic coil were 0.3 W and 0.24 W, respectively. As shown in **Figure 2B**, it was apparently observed that the integrated electromagnetic switch could provide an active cooling mechanism by rapidly reducing the heating temperature of the OG. The electromagnetic active cooling system substantially accelerated the cooling response time of OG at the different cooling intervals compared to the system without the cooling effect (B1: 50~45 $^{\circ}\text{C}$ taking 1.36 s with cooling and 1.99s without cooling; B2: 55~50 $^{\circ}\text{C}$ taking 1.30 s with cooling and 4.38 s without cooling; B3: 60~55 $^{\circ}\text{C}$ taking 1.79 s with cooling and 6.6 s without cooling) with the aid of the thermal diffusion from the thermal actuator to the magnet embedded under the platform (**Figure 2C**). While the presence of the cooling system also influenced the heating response time immensely (A3: 45~60 $^{\circ}\text{C}$ taking 2.2 s with cooling and 4.02 s without cooling) due to the heat generated from the Cu coils beneath the heating electrode for manipulating the electromagnetic field. Therefore, it is clear that applying the electromagnetic active cooling system can allow the heating electrode to achieve its target more efficiently within a short period regardless of heating and cooling conditions.

Afterward, to verify the optimal thickness of PET platforms for managing the heating and cooling systems, the OGs fabricated with distinct thicknesses of PET (25 μm , 50 μm , and 75 μm) were tested under the identical condition as the electromagnetic active cooling system performance test (**Figure 2D, E, and F**). As shown in **Figure 2D**, as the PET thickness increases, a longer response time for achieving the target temperature is displayed due to the lower heat transfer efficiency in thicker materials. Further in detail, according to **Figure 2E**, the response times of 0.6 s @25 μm , 1.09 s @50 μm , and 1.91 s @75 μm can be observed for raising the temperature from 45 to 50 $^{\circ}\text{C}$ (A1). Similarly, for the cooling condition (B1) where the temperature is decreased to the stand-by heating temperature from 50 $^{\circ}\text{C}$, 1.36 s, 1.56 s, and 2.42 s are consumed respectively. Other than the thermal ranges, A1 and B1, the whole sections are with the identical trend that thinner PET is more favorable for heat dissipation. As shown in **Figure 2F**, the thinner PET thickness induces larger temperature fluctuations when the heating temperature stabilizes at the target temperatures. Especially, the differences among the fluctuations are apparent with the target temperature of 60 $^{\circ}\text{C}$ (± 2.9 $^{\circ}\text{C}$ @25 μm , ± 2.4 $^{\circ}\text{C}$ @50 μm , and ± 1.5 $^{\circ}\text{C}$ @75 μm). Therefore, as the differences between the temperature fluctuations for the PET film with 25 μm and 50 μm are negligible while it shows considerable gaps for the response times, to compensate for both operational response and accuracy in temperature management, the OG with an optimized PET thickness of 25 μm can be concluded.

After the optimization of the PET platform, to investigate the heating power effect on the operational performance of the OGs, three different powers (0.16 W, 0.25 W, and 0.3 W) were employed on the heating electrode under the condition exploited for the prementioned tests (**Figure 2G, H, and I**). Conforming to **Figure 2G**, high-power input could generate heat quickly to reach the target temperatures, leading to a relatively rapid response time. Specifically, for the heating mode ranges which heated the thermal actuator from the stand-by temperature (45 $^{\circ}\text{C}$) to the targets (50 $^{\circ}\text{C}$, 55 $^{\circ}\text{C}$, and 60 $^{\circ}\text{C}$ respectively) under different heating power (A1: heating from 45 to 50 $^{\circ}\text{C}$ took 1.65 s @0.16 W, 1 s @0.25 W, and 0.6 s @0.3 W; A2: heating from 45 to 55 $^{\circ}\text{C}$ took 3.08 s @0.16 W, 2.2 s @0.25 W, and 1.31 s @0.3 W; A3: heating from 45 to 60 $^{\circ}\text{C}$ took 5.51 s @0.16 W, 3.14 s @0.25 W, and 2.19 s @0.3 W), the highest heating power (0.3 W) shows the apparently fastest response

time for whole sections. (**Figure 2H**). However, for the cooling mode, there are no significant differences among the heaters with different power inputs as the release of heat is strictly manipulated by the electromagnetic active cooling system. In addition, **Figure 2I** demonstrates the temperature fluctuations caused by different power inputs at different target temperatures (45 °C, 50 °C, 55 °C, and 60 °C). As the target temperatures increases, larger temperature fluctuations can be observed (at 60 °C, 2.53 °C, 2.79 °C, and 2.90 °C, respectively). It is clearly illustrated that at the target temperature of 60 °C, 0.3 W power input could induce more than 3 °C fluctuations including the error range which is critical in a term of temperature management. As a result, although the fastest response times could be achieved with 0.3 W heating power, 0.25 W was confirmed to be the optimal operational power for the thermal actuator of the OG to reduce temperature fluctuations.

Besides the optimal power for the heater, the optimal power for the electromagnetic coil was examined with three different electrical powers (0.24 W, 0.3 W, and 0.35 W). **Figure S10** exhibits the operational performance of the OGs as a function of the power into the electromagnetic coil including the response time (**Figure S10A and B**) and temperature fluctuation (**Figure S10C**). Referring to **Figure S10A and B**, both heating and cooling response times (from 45 to 60 °C and from 60 to 45 °C, respectively) showed insignificant differences for the different electrical powers (for heating, 2.19 s @0.24W, 2.07 s @0.3 W, and 2.01 s @0.35 W and for cooling, 1.79 s @0.24W, 2.07 s @0.3 W, and 1.65 s @0.35 W). Furthermore, for the temperature fluctuations, the effect of the power into the coil were unremarkable as well (0.24 W @1.94 °C, 1.42 °C @0.3 W, and 1.50 °C @0.35 W) (**Figure S10C**). Then, as shown in **Figure S11**, with the increment in the electrical current applied to the coil, the lifting force generated by the electromagnetic field was influenced to be stronger and consequently, the lifting height was increased (0.19 cm, 0.21 cm, and 0.24 cm respectively with the increasing power). Although the electrical power was raised from 0.24 to 0.35 W, the height was slightly increased which was hard to expect any consequential effect. Thus, 0.24 W was selected for the coil to save the systemic power of the whole device. Following, the thickness of the paraffine wax layer that is the source of the smell has been investigated with different wax thicknesses (50 μm , 200 μm , and 400 μm) (**Figure S12**). According to **Figure S12A**, it is clearly displayed that the thinner paraffin wax could significantly reduce the

response time to the target temperature during the heating and cooling due to the high specific heat capacity of the wax. As a result, increasing the thickness of paraffin from 50 μm to 400 μm significantly increased the temperature response times of the OG from 2.2 s to 6.8 s during the heating phase (A3, 45~60°C) and from 1.8 s to 9.4 s during the cooling stage (B3, 60~45°C) (**Figure S12B**).

Following, for examining the temperature managing capacity of the OGs at different environmental temperatures (20 °C, 25 °C, and 30 °C), an OG was operated with heating and coil powers of 0.16 W and 0.25 W, respectively (**Figure S13**). To conduct the experiment, the heating electrode of the OG was manipulated under the identical condition to the prementioned tests, which was heated from the stand-by heating temperature (45 °C) to three different targets (50 °C, 55 °C, and 60 °C) (**Figure S13A**). As illustrated in **Figure S13B**, shorter heating response times were observed for the OG for the whole heating ranges (from 45 to 50 °C, from 45 to 55 °C, and from 45 to 60 °C) as the ambient temperature was increased from 20 to 30 °C correspondingly (comparing the heating responses at the environmental temperatures of 20 °C and 30 °C, from 45 to 50 °C: 0.671 s took more for 20 °C, from 45 to 55 °C: 1.135 s took more for 20 °C, and from 45 to 60 °C: 0.89 s took more for 20 °C). Consequently, it was demonstrated that the ambient temperature influences the response time of the OGs.

With the optimized response time and input power of the OGs, we next studied the smell-generating capacity of the OGs (**Figure 2J**). During the testing, a commercial ethanol sensor (TGS823) was fixed 1 cm above a working OG and while continuously monitoring the nearby ethanol concentration in ppm by reading the voltage variation of a 5 k Ω resistor connected in series. By switching the heating temperature of the OG from 45 to 50 °C, to 55 °C, and finally, to 60 °C, the ambient ethanol concentrations could reach up to 180 ppm, 261 ppm, and 565 ppm, respectively. This result demonstrates that higher heating temperatures contribute to a higher generated odor concentration and also, the magnitude of the olfaction interface could be manipulated by the heater temperature. In addition, the operational stability and repeatability of the OG were proved through a cyclic test (**Figure 2K**). In this experiment, the OG was controlled to be stabilized at 45 °C and 50 °C in a cycle with the frequency of 0.1 Hz for more than 20 hr. As a result, the stability and repeatability of the OG were

successfully proved as the OG generated a continuous and repeated operational pattern according to the experimental design during the iteration test (**Figure S14**). Lastly, to reveal the relationship between the smell-generating time and the paraffin wax mass, the duration of odor release of eight different odorants with different paraffin wax masses (2.5 mg, 10 mg, and 20 mg) was recorded at the heating temperature of 60 °C. In this experiment, although a higher portion for the perfume increased the smell-generating time,⁴⁶ the blend ratio between the wax and perfume was optimized to be 10:1 among 10:1, 10:2, and 10:3. This is because, as shown in **Figure 2L**, whole eight smell types with 20 mg of paraffin wax (wax: perfume = 10:1) could generate more than 3 hr, including error ranges which are sufficient for covering the optimum operation time. Furthermore, as indicated in **Figure 2L**, increasing the paraffin mass significantly extended the release duration for all sets of smell. In addition, among them, the fifth odor had the longest releasing time (2 hr, 6.3 hr, and 11.67 hr respectively) while the sixth odor showed the shortest ones which were 0.5 hr, 0.83 hr, and 2.5 hr correspondingly. Therefore, it was proved that the amount of paraffin highly affects the smell-generating time and at the same time, the material for the smell also influences the duration of the smell immensely. Consequently, through experiments, the temperature-sensing capacity of the OG was confirmed with linearity while the effect of the electromagnetic active cooling system was illustrated as well. Furthermore, the structural dimension which is the thickness of the PET platform and electrical power input for both heater and Cu coil had been optimized (25 μm, 0.25 W, and 0.24 W respectively) for achieving rapid thermal management of the heater. Additionally, the relationship between the thickness and mass of paraffin wax and smell-generating was demonstrated that the thinnest wax layer (50 μm and 2.5 mg) showed the fastest response time while the higher mass (200 μm and 20 mg) of the wax was with the longest smell-generating functionality. Then, the operational stability and repeatability of the OG were approved through the cyclic test for over 20 hr. Through the operational performance tests and electrical power and structural dimension optimization of the OG, the superiority of the reporting OG over the reported or commercialized olfaction interfaces was certainly revealed (**Table S1**).^{38, 46-52} Compared to the reported smell-generating devices, the OG was apparently lightweight, small, wirelessly operated, and flexible, which is suitable for skin-integrated applications. At the same time, it

presented a relatively rapid response time and a wide variety of odor options (24 in total) for achieving fast reactions against the stimuli and high smell-generating efficiency.

The wearable olfaction interface system embedded with eight OGs for enabling deafblind users who are with physical barriers in vision and audio to wirelessly communicate with non-deafblind people through a graphical user interface (GUI) was introduced (**Figure 3A**). To deliver external information, the sense of smell was exploited to convey information in a more natural, accessible, and undisturbed manner while a self-developed GUI installed in an external handheld electronics was used to control the device wirelessly for generating types of odors indicating specific messages (**Figure S15**). **Figure 3B** illustrates the mechanism for sensation and perception of olfaction. The sense of olfaction is induced by smell entering the nasal cavity through inhalation and interacting with receptors in the olfactory epithelium to elicit electrical signals which are then transmitted signals with chemical information to the olfactory bulbs and olfactory cortex of the brain where used to process and percept of odors.⁵³ Unlike other senses, the olfactory system has a neural overlay on the amygdala and is directly connected with many brain areas including the hippocampus and hypothalamus which are involved in processing emotions and memory formation.^{54, 55} Therefore, odors can ignite memories, trigger emotions, and influence a wide range of behavior. To ensure the operational safety of the device, the thermal image of the device after 10 mins of operation in air at the OG target temperature of 60 °C had been recorded. As shown in **Figure 3C**, the temperature of the device was successfully maintained in a relatively low range (22 ~ 37 °C) that would not cause any skin burn that could result from thermal application more than 44 °C for a few hours and even a few seconds at above 70 °C to the users.⁵⁶⁻⁵⁸ In addition, to further identify its thermal safety in practical applications, the operating device was directly placed on the epidermis between the lip and nose to monitor the target skin temperature (**Figure S16A**). As illustrated in **Figure S16B**, the distance between the nearest OG unit where a heating electrode is installed and the nose was measured to be 8 mm, that is sufficiently far to be out of the heating effect range. Furthermore, as the heaters in the OGs are physically separated from the main body of the device by shells, heat conduction from the heating electrodes to the skin is completely prevented. Additionally, to further verify the biosafety of the olfaction device, the thermal effect of

the device on the target skin area where the device is directly mounted was investigated. For measurement, the operating device was removed from the target after the specific application times (1 min, 3 min, 5 min, 7 min, 9 min, and 11 min), and then, thermal images were recorded to investigate the thermal distribution of the working device on the skin (**Figure S17**). As a result, the temperatures of the target skin area were measured to be under 40 °C for all operation times, which is safe for long-term application.

Following, the working mechanism of the wireless application of the wearable olfaction interface system was exhibited (**Figure 3D**). As the whole device is composed of two symmetrical FPCBs installed with four OGs separately, two identical Bluetooth modules were adopted for transmitting the data of the thermal information of each OG to the external device for GUI display and receiving the command for operating specific smell from the GUI by pressing the buttons (**Supplementary Movie 1**). Therefore, as shown in **Supplementary Movie 2**, the smell-generating functionality could be conducted while the thermal information for each OG was clearly monitored through eight different graphs in real-time with the timeline in seconds by manipulating the GUI.

Furthermore, the maximum distances for wireless transmission at different heights (0 m, 0.5 m, 1 m, and 1.5 m respectively) and sender and receiver arrangements (face-to-face, face-to-back, face upward, and back-to-back modes) were investigated to discover the performance of the wireless communication between the device and external GUI installed electronics. According to **Figure 3E**, it showed a trend that as the height difference increased, the transmission lengths had been decreased with considerable linearity (10.93 m, 9.37 m, 8.2 m, and 6.83 m correspondingly) while the Bluetooth modules were facing each other. Then, conforming to **Figure 3F**, it was revealed that the orientations of the sender and receiver would highly affect the transmission distance of the system (10.93 m, 3.67 m, 6.5 m, and 1.7 m sequentially). The face-to-face orientation between the user and receiver displayed the longest average maximum transmission distance while the back-to-back orientation showed the shortest transmission distance. Therefore, through the investigation of the maximum distances for wireless transmission, it was verified that the wireless olfactory system can provide real-time wireless communication while the free movement of the users is guaranteed. Afterward, the continuous

operating time of the system was examined with a commercial lithium-ion battery (7.4 V, 1600 mAh) for measuring the practical working duration (**Figure 3G**). In the test, 5.3 V was lastly recorded as the power source as the electrical power less than 5.3 V was unsuitable for establishing rapid response time for the temperature management. As a result, the system could maintain regular operation for more than 4 hr which is sufficient for practical application in daily life. In addition, to specifically measure the operational durations at different working conditions, different numbers of the OGs were stabilized at 60 °C with the battery exploited in the aforementioned experiment while the others were maintained at the stand-by temperature (45 °C) (**Figure S18**). As a result, the five different sets of the operating time (0, 2, 4, 6, and 8 OGs at 60 °C) were recorded (**Figure S18A**), and according to **Figure S18B**, it was observed that as the number of the OGs stabilizing at 60 °C was increased, the operation time was decreased correspondingly (2.72 hr, 2.41 hr, 2.23 hr, 2.20 hr, and 2.00 hr). As a result, by considering **Figure 3G** and **S18**, the optimum operation time was determined to be 3 hr. Consequently, after testing the practical application, the operational performance of the wireless olfactory interface system that is composed of the smell-generating device and external electronics for GUI display was verified as the device auspiciously generated the specific odors indicating corresponding messages with the buttons pressed on the GUI side while the real-time thermal data of the OGs were exhibited. Furthermore, by analyzing the maximum transmission lengths for wireless communication between the device and an external device at different heights and orientations and the practical working time, it was proved that the system is available for daily life usage.

Lastly, two communication methods between a deafblind patient and a medical doctor which are the simple touch-based hand-in-hand and smell message delivery by the olfaction system were compared to demonstrate the advantages of the olfaction interface for message delivery system over the touch-based approach (**Figure 4A**). Touch-based message delivery is one of the important communication methods for deafblind individuals.^{59, 60} However, in many cases, such as in hospitals and other public places, a large number of people are not trained in sign language or Braille reading due to their learning difficulties and lengthy training duration¹⁴⁻¹⁸ and therefore, causing barriers to information exchange hand-in-hand. Meanwhile, the hand-to-

hand transmission of information can result in contact contamination among people which increases the risk of bacterial or viral cross-infection.^{61, 62} Compared to the conventional communication method, the wireless olfaction system can provide a message delivery system wirelessly among individuals in non-contact ways. Therefore, during a pandemic such as COVID-19, an olfaction message delivery system can benefit users. With simple training in advance, the wireless olfaction system can express and transmit certain information in a straightforward, rapid, and efficient way, especially for complex and lengthy messages like, “You are recommended to be hospitalized for further treatment.”. However, on the other hand, even though it was reported that 1 trillion odors can be detected by human olfaction sense,⁶³ still, there is a limitation in the wireless olfaction interface. Due to the olfactory sensing and odor molecular characteristics, there are a large number of overlapping sets of odorants which hinders discriminating similar smells precisely.^{22,33} Therefore, although the device can employ extended numbers of the OGs by further miniaturizing the OG unit and applying multiple layers for the OG platform to provide a wider variety of messages, the number of single odorants that are specifically with different smell variations is limited. To briefly introduce the smell-generating mechanism of 24 types of odors, six different odors conveying separate messages were generated by exploiting eight OGs by heating one or two OGs to 55 °C. As shown in **Figure 4B**, single or multiple OGs could be controlled to generate specific smell according to the order from the GUI while the remaining OGs were stabilized at 45 °C. In total, by turning on 8 OGs separately and randomly mixing 2 odors generated by separate 2 OGs from left and right olfaction FPCBs each, the olfaction system could support 24 different messages in the form of smell. Following, the average delivery durations and recognition rates of 24 messages exploiting the olfaction interface and hand-in-hand method were measured to compare their communicational performance and efficiency.

As shown in **Figure 4C**, it is apparently illustrated that the delivery times for the hand-in-hand method were relatively lengthy compared to the wireless olfaction interface system that the delivery time was fixed at 10.99 s which was obtained by averaging the time taken for the OGs to sufficiently release the smell for sensing. For the hand-in-hand method, the delivery durations were highly dependent on the characters of the messages, and therefore, for the longest one, it took 71.33 s to successfully convey the message while the shortest one took

26.67 s which was still more than double the olfaction interface system. Furthermore, **Figure 4D** shows the recognition rates of hand-in-hand and olfaction methods for each message with 10 volunteers involved. The recognition rate of the hand-in-hand is generally lower than that of the olfactory process for the 24 messages demonstrating that the olfaction system exhibits higher message delivery efficiency than that of the hand-in-hand method. Especially, for message 11, none of the volunteers interpreted the message correctly through the hand-in-hand method while the olfaction interface system showed a 70% of recognition rate. As a result, through comparison, the superiority of the olfaction interface system over the touch-based method was proved with rapid delivery duration and advanced recognition rate.

Then, to investigate the effect of the training on the smell recognition rate, 31 volunteers, organized by 17 females and 14 males, were trained with 24 types of odors indicating different messages for 0.25 hr and 24 hr, respectively, before the recognition tests. As shown in **Figure 4E**, after 24 hr of the training, a higher recognition rate (64%) was observed while when the testers were trained with 0.25 hr, they only recognized 45% of the testing sets. Specifically, there was a wide range of individual variations among the volunteers (from 0.125 to 1); however, after 24 hr of training, most volunteers (24 out of 31) showed a relatively higher recognition rate than training for 0.25 hr, while for the ones who showed decreased recognition rates, the differences between 0.25 hr and 24 hr training were comparably small (**Figure 4F**). Therefore, the recognition accuracy can be improved immensely when a longer period of training was adopted for the olfaction interface communication. Afterward, the relationship between recognition rate and gender was examined by testing 17 female and 14 male volunteers with 24 types of odor. According to **Figure 4G**, it could be observed that the women group was with a higher recognition rate of 66%, while the men group recognized 61% of the testing sets. In addition, in most cases, females showed higher recognition rates compared to males. However, for message number 9, 18, and 24, men presented exceptional recognition rates with more than 10% differences. As a result, it was verified that women are more sensitive to smell^{64, 65} and on the other hand, there are specific odors that are more sensitive to men.

After examining the effect of the training on the smell recognition rate, to evaluate the message-delivering performance of the olfaction interface in practical applications, the odor intensity at different working OG temperatures for investigating users' perception of the odors at different smell-generating temperatures and the odor retention time after shutting down the odor for identifying the duration required to discontinue the specific smell for eight separate odors were assessed by testing 19 volunteers, composed of 11 females and 8 males (**Figure S19** and **S20**). As the intensity of releasing odor is considered to be another factor that users judge when delivering odor-based messages, to discover the heating effect on the odor intensity, an odor intensity score ranging from 0 to 5 (0: Nothing, 1: Little, 2: Some, 3: Moderate, 4: Little intense, and 5: Too strong) was designed. Then, based on the intensity score, 19 volunteers were tested with the eight different odors generated at 45 °C and 50 °C separately, which are the stand-by and the minimum working temperatures of the olfaction interface. As a result, higher odor intensity scores were achieved at 50 °C with an average intensity score of 2.51 (1.37, 1.42, 2, 1.79, 1.53, 1.26, 1.84, and 1.26, respectively from odor 1 to 8), compared to 1.56 at 45 °C (2.32, 2.74, 2.84, 2.58, 1.95, 2.26, 3.32, 2.11, correspondingly) (**Figure S19A**). Furthermore, regardless of gender, obvious intensity differences between the applied temperatures at 45 °C and 50 °C were observed for both female and male groups (**Figure S19B** and **C**), while depending on the smell type, volunteers sensed different levels of smell, like for odor 7, relatively higher intensity score was recorded at 50 °C. Following, for the odor retention time measurement, the smell was generated by melting paraffin wax at 50 °C, and then, when the volunteers sensed the specific odor, the heater was cooled down to the stand-by temperature to terminate the release of the smell. As illustrated in **Figure S20A**, the average odor retention time was measured to be 24 s with an average error range of ± 16 s (22 s, 25 s, 22 s, 24 s, 25 s, 25 s, 25 s, and 26 s, respectively from odor 1 to 8). Furthermore, in detail, similar to the previous experiment, the acceptable ranges of the average odor retention time for both women and men groups were observed which were 22 s (23 s, 26 s, 27 s, 24 s, 28 s, 30 s, 26 s, and 30 s) and 27 s (21 s, 23 s, 19 s, 24 s, 23 s, 21 s, 25 s, and 23 s) respectively with the identical average error range of ± 16 s (**Figure S20B** and **C**).

Next, the recognition accuracy of a single and the combination of two odors had been compared by testing 31 volunteers for determining the maximum number of smells that could be mixed (**Figure 4H, I, and J**). For the experiments, the recognition rates of a single odor from a specific OG were compared to the average of the mixed smell generated from the combination of the specific OG and an OG placed in the other olfaction FPCB (*e.g.*, for the single odor number 1, the combination of OG1 and OG5, OG6, OG7, and OG8, respectively). During the tests, although the maximum number of messages that could be generated was 248 ($C_8^1 + C_8^2 + C_8^3 + C_8^4 + C_8^5 + C_8^6 + C_8^7 + C_8^8$), it was limited to 24, which was the combination of 8 single odors and 16 two odors blended smells. As illustrated in **Figure 4H and I**, obvious drops in the recognition rates were observed for two mixed smells compared to the single odors, and therefore, to further increase the number of choices for the smells, it is expected to design a specialized mask for trapping the odors for an extended time. As indicated in **Figure 4H**, the recognition rates of the combined odorants were lower than that of a single odorant for eight different odorants in all cases. Furthermore, except for the ninth and twenty-seventh volunteers, higher recognition rates were measured for the single odorants (**Figure 4I**). Therefore, similar to the recognition test with different training times (**Figure 4F**), individual variations among the volunteers were observed. Finally, the volunteers were trained with 24 types of smell for 0.25 hr and 24 hr respectively to analyze the effect of the training time on the perception of the combined odorants. As shown in **Figure 4J**, the recognition accuracies for both single and two mixed odors immensely increased after 24 hr of training (0.78 and 0.57 correspondingly) compared to the 0.25 hr trained results which were 56% and 40% respectively. Furthermore, especially for the combined odorants, it showed a 57% of recognition rate which is adequate for application after 24 hr of training. As a result, the maximum number of the combination of smells was confirmed to be two as if more than three odors are mixed, the recognition rate is expected to be below 50% which is not sufficient for the applications. Through the experiments, the superiority of the olfaction interface system over the touch-based hand-in-hand method had been verified from the perspectives of communicational performance and efficiency. In addition, it was clearly shown that longer training time contributes to improving recognition accuracy as the majority resulted in higher recognition rates even though there were individual variations.

Finally, the maximum number of the mixing smell was optimized to two due to the decreasing recognition rate with increasing the number of combinations.

Conclusion

In summary, a skin-integrated and wireless olfactory interface system exploiting arrays of eight miniaturized OGs for creating 24 different smells representing specific messages by operating one or two OGs synchronously has been introduced. The smell-generating device that is attached to the philtrum of the user is wirelessly linked with the customized GUI on an external device for controlling the device by pressing the buttons and displaying the thermal data of the OGs simultaneously. Therefore, by manipulating the GUI, ordinary people can deliver messages from a remote distance to the deafblind who conventionally obtain information relying on touch sensation. To achieve a miniaturized and simplified structure of OGs, the heating electrode is used as a thermal source and sensor concurrently with the aid of the circuit design while the temperature of OGs is rapidly regulated and stabilized by switching high and low power lines and the electromagnetic active cooling system. Through experiments, the structural dimension that is the thickness of the PET platform and the electrical power for heaters and Cu coils were optimized for establishing enhanced thermal management. Finally, by comparing the recognition rates of messages delivered by the olfaction interface system and touch-based hand-in-hand method, the advantages of the smell message delivery system are illustrated in the aspects of performance and duration. As a result, the potentiality of the olfaction interface system as an alternative message delivery method targeting deafblind patients has been exhibited.

Experimental Section

Fabrication of OGs. The fabrication firstly started with cleaning a quartz glass using acetone, alcohol, and deionized water (DI water) in sequence. A thin layer of polyimide (PI) with a thickness of 25 μm was fixed onto the glass sheet, then Au/Cr (200/40 nm) were deposited onto the PI film by magnetron sputtering using a machine (Q150TS, QUORUM). Through photolithography and etching, the metal patterns in the desired geometries could be obtained on the PI film. For the photolithography, a positive photoresist (PR, AZ 5214, AZ Electronic Materials) was spin-coated at 3000 rpm for 30 s, soft-baked on a hot plate at 110 $^{\circ}\text{C}$ for 4 min, and then, exposed to ultraviolet light for 5 s. After that, the photoresist coated was developed by a developer (AZ 300 MIF) for 15 s and then, by using acetone and DI water, the PR was rinsed away. After that, spin casting a layer of PI (2 μm , 3000 rpm for 30 s, annealed at 250 $^{\circ}\text{C}$ for 30 min) and then selectively etched by Oxford Plasma-Therm 790 RIE system (patterns defined by photolithography similar as previous step) at the power of 200 W for 10 min, formed encapsulation layers for serpentine-designed Au/Cr traces while exposing the connecting patches for later wired connections. After rinsing the PR away, we could obtain the heating electrode. Fix the heating electrode onto a cantilever-like PET film (25 μm) with a copper coil mounted at the bottom. Following, stick the sample onto a prepared PET platform fabricated by a laser cutting machine. Then, fix the sample onto a chamber with a magnet embedded in the middle, where the back cover is a flat PET film (25 μm), and the supporting wall is the flexible PDMS ring. Finally, a 3D-printed shell with the melting point of 70 $^{\circ}\text{C}$ is attached onto the top of the heating electrode.

Fabrication of the olfaction interface. To fabricate the olfaction interfaces, FPCBs were manufactured by commercial printed circuit board processing technology with two copper conductive layers covered with polyimide (PI, thickness – 270 μm). On the top and bottom layers of FPCBs, circular and rectangular Cu patches were exposed to place the electrical components and OGs according to the design while the electrically conductive pins of the components were soldered with a low-temperature soldering paste (LF999, KELL YSHUN Technology Co. LTD). A number of electrical components including the microcontroller unit (ATMEGA328P-MU), capacitors, resistors, crystal oscillator (16 MHz), two N-type MOSFETs, two P-type MOSFETs, decoders (SN74LVC1G19DBVR), four LDOs with output values of 5.6 V, 5 V, 3.3 V, and 2.6 V respectively, 24-bit ADC module (ADS1220IRVAR), LEDs, Bluetooth module (WH-BLE103), and OGs were involved to operate a FPCB. To fully operate eight OGs in total, two symmetrical FPCBs were exploited as a set. After components integration for both FPCBs, the soldered FPCBs were placed in a customized mold and poured with PDMS (crosslink: PDMS = 1:20, 145 kPa) for encapsulation. Then, the device was cured at 110 $^{\circ}\text{C}$ for 5 min. As a result, the skin-integrated olfaction interface with eight OGs was fabricated for delivering messages to deafblind patients by generating specific odors.

Operation of the olfaction interface in message delivery. For operating the olfaction interface in message delivery, two symmetric FPCBs are exploited. For each FPCB, four OGs are controlled with a MCU (ATMEGA328P-MU, Microchip Technology Inc.) by switching the high and low power lines for each heating electrode through two different load switches composed of N-type MOSFET (NMOS) and P-type MOSFET (PMOS) respectively. For the high power line which is turned on to heat up the heater when the heating temperature is lower than the target temperature, NMOS (SI2336DS-T1-GE3, Vishay Intertechnology, Inc.) and PMOS (SI2307CDS-T1-GE3, Vishay Intertechnology, Inc.) are used to enable the flow of 5.6 V from a LDO (LP38690SDX-ADJ/NOPB, Texas Instruments Inc.). On the other hand, for the low power which is turned on to cool down the heater when the heating temperature is higher than the target temperature, NMOS (SI2312BDS-T1-GE3, Vishay Intertechnology, Inc.) and PMOS (SI2301BDS-T1-E3, Vishay Intertechnology, Inc.) are used to allow the electrical flow from 2.6 V LDO (MIC5353-2.6YMT-TR, Microchip Technology Inc.). As 2.6 V is sufficiently high to generate heat on the heater, the SMD type of LED is attached at the end of the low power line to reduce the voltage to 0.6 V while the LED prevents any backflow of the current and electrical interruption between the high and low power lines. Then, as the heating electrode is connected to a reference resistor by forming a voltage divider circuit while the output voltage pin is connected to a 24-bit ADC module (ADS1220IRVAR, Texas Instruments Inc.) that is connected to the MCU through serial peripheral interface (SPI) for data transmission, the different output voltage is sensed in high resolution by the ADC module according to the temperature of the heating electrode due to the resistance change. As the reference voltage of the ADC module is 1.04 V while it is with 20-bit effective resolution (range from 0 to 1048575), the resistance of the heating electrode can be calculated back from the resulted voltage and converted into thermal data in the MCU more accurately. Furthermore, due to the presence of the multiplexing function in the chip, all four heating electrodes on the single FPCB can be monitored simultaneously. As a result, by switching the two digital pins of the MCU contrastively, the heating temperature of an OG can be controlled and monitored simultaneously without any delay and disconnection. Furthermore, for managing the electromagnetic active cooling system of the OG that is composed of a magnet and a coil, a decoder (SN74LVC1G19DBVR, Texas Instruments Inc.) connected with the ends of the coil generate contrary voltage outputs (2.5 and -2.5 V respectively) from two pins and vice versa accordance with the digital signal from the MCU (high or low). Therefore, according to the applied digital signal, the direction of the electrical current produced by the decoder shifts for lifting up and down the heating platform of the OG. Consequently, three digital GPIOs of the MCU are exploited to operate a single OG while four OGs are manipulated on each FPCB. Following, the whole system which is the combination of the two symmetric FPCBs installed with eight OGs in total is powered by a commercial lithium-ion battery (7.4 V, 1600 mAH). Then, the main power passes through four different LDOs on each FPCB for stepping down 7.4 V into 5.6 V, 2.6 V, 5 V (LP38692SDX-5.0/NOPB, Texas Instruments Inc.), and 3.3 V (TPS76933DBVR(PB-FREE), Texas Instruments Inc.) respectively. As prementioned, 5.6 and

2.6 V LDOs are installed for managing the high and low power lines and the electromagnetic active cooling system of the OGs while 5 and 3.3 V LDOs supply the operating power for the MCUs and Bluetooth modules (WH-BLE106, Jinan USR IOT Technology Limited) correspondingly. Thus, through the Bluetooth modules embedded in each FPCB, the commands for the odor-generating patterns are transmitted to the MCU for producing a specific smell for message delivery after the corresponding button is pressed in GUI. Simultaneously, the temperatures of each OG are transmitted to the external paired computer for GUI display. Therefore, to generate a smell from an OG, the high power line is turned on and applied to the heating electrode to rapidly increase the temperature to its target temperature (higher than the melting point of the paraffine wax, approximately 50 °C) from its stand-by heating temperature (45 °C) while the decoder generates a forward direction of current flow into the coil to lift up the heating platform. As a result, the wax is melted down and the odor is diffused out from the OG. Furthermore, once it reached the target temperature, the current temperature and the target temperature are compared for every systemic loop and the high and low power lines are turned on and off accordingly to stabilize the temperature of the OG at the target until the command is received. Additionally, to terminate odor generation, a reverse direction of current is generated by the decoder to activate the cooling system. Subsequently, the heating platform is contacted with the magnet to release heat quickly and when the temperature of the OG drops to the stand-by temperature, the heating platform is lifted up again by switching the direction of current into the coil. Then, the temperature is stabilized at the stand-by temperature until the further command input.

Mechanical simulation. The FEA commercial software ABAQUS (Analysis User's Manual 2020) was used to calculate the bent and twisted of structure. The epoxy, wax, PI, adhesive and PET layers were modelled by 1.39 million linear hexahedral elements of type C3D8R. Heating electrode layer were modelled by 0.12 million linear quadrilateral elements of type S4R. The minimal element size was 0.001 mm, which ensured the convergence and the accuracy of the simulation results. The elastic modulus (E) and Poisson's ratio (ν) used in the analysis were $E_{\text{Epoxy}}=3.35$ GPa, $\nu_{\text{Epoxy}}=0.35$, $E_{\text{wax}}=0.8$ GPa, $\nu_{\text{wax}}=0.48$, $E_{\text{PI}}=3.15$ GPa, $\nu_{\text{PI}}=0.34$, $E_{\text{Adhesive}}=1$ GPa, $\nu_{\text{Adhesive}}=0.48$, $E_{\text{PET}}=3.5$ GPa, $\nu_{\text{PET}}=0.35$, $E_{\text{Electrode}}=79$ GPa and $\nu_{\text{Electrode}}=0.41$.

Data availability

All data are contained within the manuscript. Raw data are available from the corresponding authors upon reasonable request.

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Associated Content

Supporting Information

The Supporting Information is available free of charge at (**Address**).

Supplementary Information section: Comparison table among the reported/commercialized olfaction interfaces with the introducing system, overview of the assembling components for the olfaction interface system, mechanical stimulation result of the OG PET film and heating electrode, operation logic of the olfaction interface, thermal distribution of the OG at different heating temperatures, skin adhesive property of the device, operational performance of the OG under different electromagnetic coil power, wax thicknesses, and environmental temperatures, detailed operational response of the OG, optical image of the self-developed GUI for wirelessly controlling the system, skin thermal safety experimental setup and resulted thermal images of the target skin, operational durations of the system under different working OG units, odor intensity differences at different heater stabilizing temperatures, and odor retention time differences (**DOC**)

Supplementary Movie 1: GUI of the operating wirelessly controllable olfaction interface system (**MP4**)

Supplementary Movie 2: Real-time operation of the wireless olfaction interface system (**MP4**)

Author contributions

W.P., Y.M.L., Y.J., and R.S. contributed equally to this work. W.P., Y.M.L., and X.Y. conceived the ideas, and designed the experiments. W.P., Y.M.L., Z.Z., Y.H.L., and X.Y. wrote the manuscript. Y.M.L., X.H., W.P., Z.Z., R.S., Y.J., J.N., Y.C., Y.G., D.L., Q.Z., Z.G., Z.D., G.Z., S.J., W.S., W.P., Y.Y., and Y.W., performed experiments and analyzed the experimental data. W.P., and C.Y. performed electrical circuit designs. Z.Z. and Y.H.L. performed the theoretical mechanical analysis.

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Figure captions

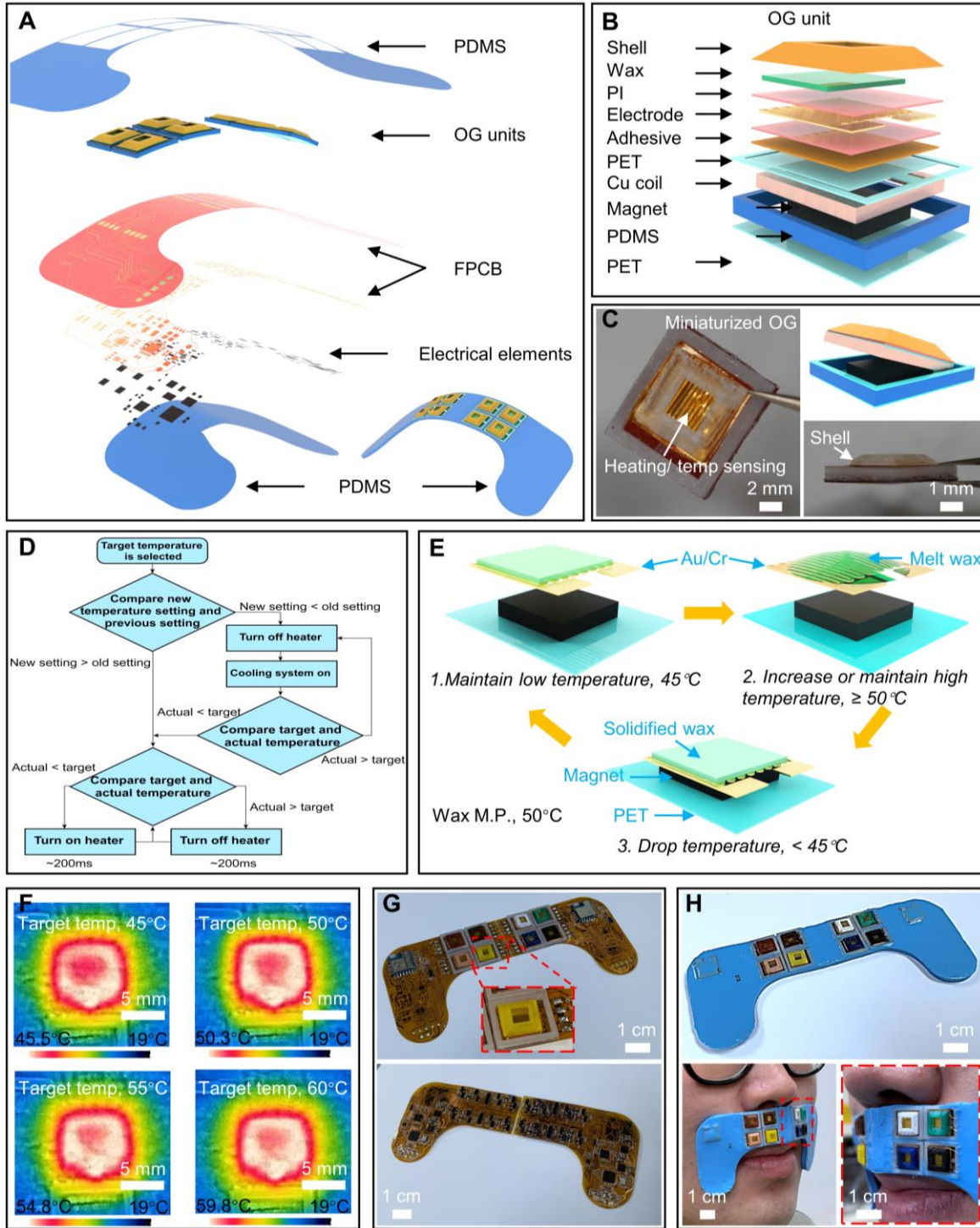


Figure 1. Overview of the skin-integrated wireless odor message delivery electronics system (A) Exploded view of the wirelessly controlled olfaction interface. (B) Schematic illustration and (C) optical images of the OG unit. (D) Closed-loop operational Logical flow of the OGs. (E) Procedures for olfactory control of the OGs. (F) Thermal distribution of the OGs with the target temperatures varying from 45 to 60 °C. (G) Optical image of olfaction interface with FPCB and eight OG units without PDMS encapsulation. (H) Optical image of olfaction interface of eight OG units mounted onto human upper lip.

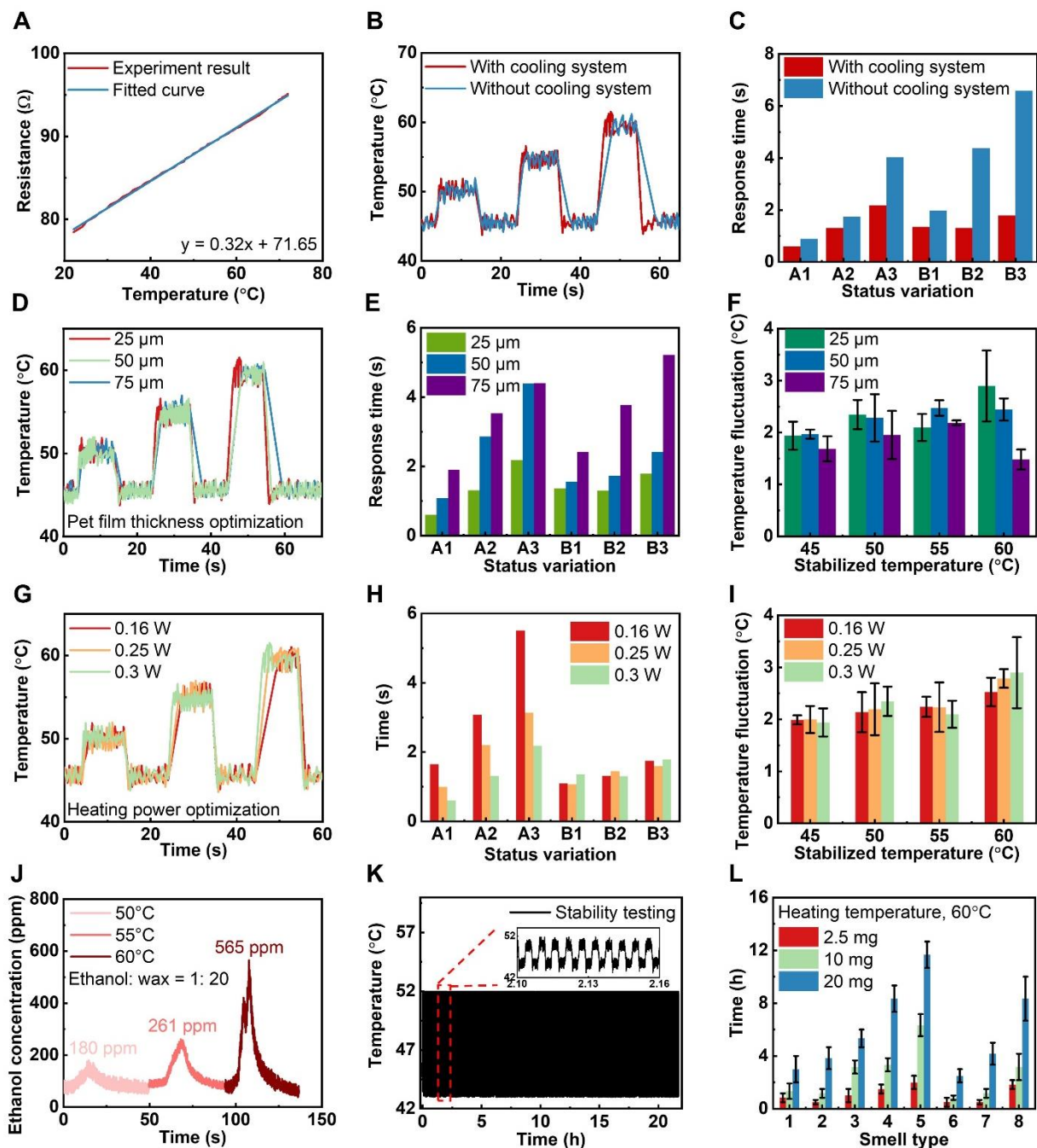


Figure 2. Experimental results exhibiting electrical and operational performance of the olfaction system. (A) Electrical resistance of the heating electrode of OGs as a function of its temperature with fitted curve line. Temperature variations of the OG with/without a cooling system while the heating temperature is switched between 45 and 50, 55, and 60 $^{\circ}\text{C}$ (B) and the corresponding operational response times (A1: 45~50 $^{\circ}\text{C}$, A2: 50~55 $^{\circ}\text{C}$, A3: 55~60 $^{\circ}\text{C}$, B1: 50~45 $^{\circ}\text{C}$, B2: 55~50 $^{\circ}\text{C}$, B3: 60~55 $^{\circ}\text{C}$) (C). Operational performance of the OG as a function of the thickness of PET with 25, 50, and 75 μm while the heating temperature is switched between 45 and 50, 55, and 60 $^{\circ}\text{C}$, respectively (D), with the analyzed results displaying response time (A1: 45~50 $^{\circ}\text{C}$, A2: 50~55 $^{\circ}\text{C}$, A3: 55~60 $^{\circ}\text{C}$, B1: 50~45 $^{\circ}\text{C}$, B2: 55~50 $^{\circ}\text{C}$, B3: 60~55 $^{\circ}\text{C}$) (E), and temperature fluctuation (F). Operational performance of the OG at the different power inputs (0.16, 0.25, and 0.3 W) to the electromagnet Cu coil of the OG while the heating temperature is switched between 45 and 50, 55, and 60 $^{\circ}\text{C}$, respectively (G), with the analyzed results displaying response time (A1: 45~50 $^{\circ}\text{C}$, A2: 50~55 $^{\circ}\text{C}$, A3: 55~60 $^{\circ}\text{C}$, B1: 50~45 $^{\circ}\text{C}$, B2: 55~50 $^{\circ}\text{C}$, B3: 60~55 $^{\circ}\text{C}$) (H), and temperature fluctuation (I). (J) Odor generating performance of the OG at the different heating temperatures (50, 55, and 60 $^{\circ}\text{C}$). (K) Cyclic performance of the OG over 20 hr. (L) Odor generating time of the OG with eight different odorants at the different paraffine wax masses of 2.5 mg, 10 mg, and 20 mg.

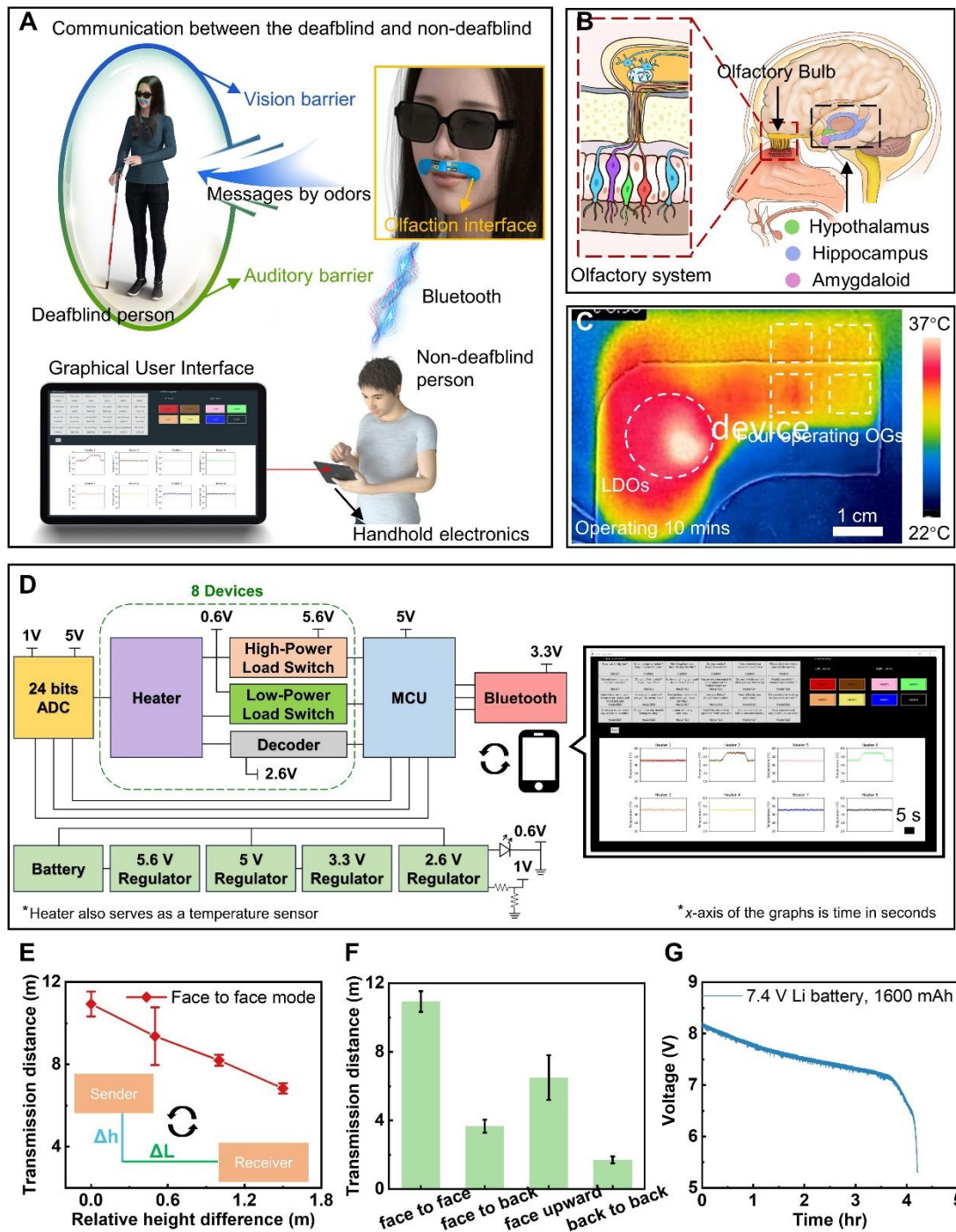


Figure 3. Overview of the practical application and electrical responses of the skin-integrated wireless odor message delivery electronics system. (A) Schematic illustration of the practical application for wearable olfaction interface. (B) Illustration of sensation and perception of olfaction. (C) Thermal image of olfaction interface after working 10 mins. (D) Block diagram of the skin-integrated wireless odor message delivery electronics system and the corresponding customized GUI. (E) Maximum transmission distances (ΔL) between sender and receiver at different heights (ΔH). (F) Maximum transmission distances at different sender-receiver orientations. (G) Operational duration of the olfaction interface with the commercial rechargeable Li battery (7.4V, 1600mAh).

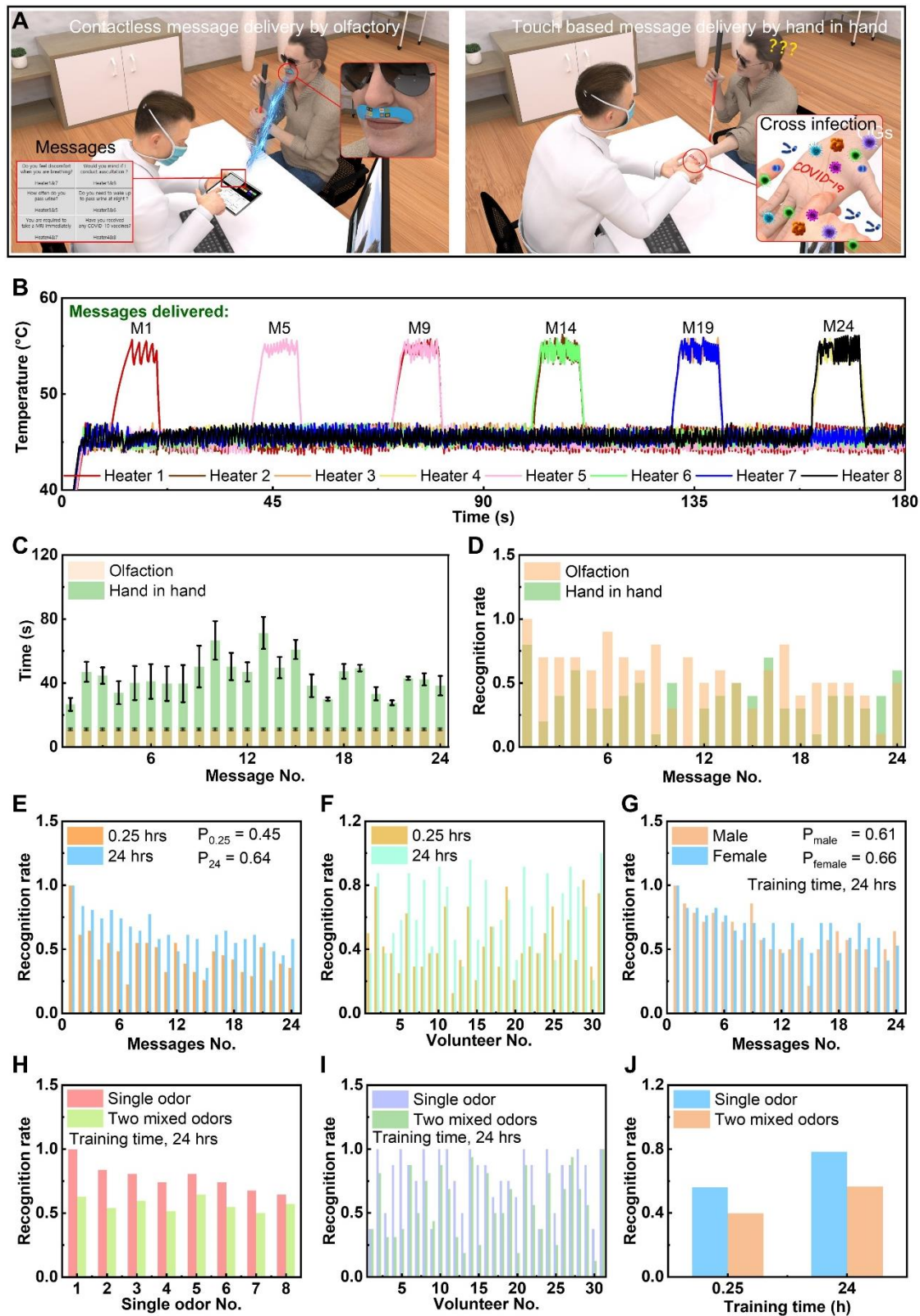


Figure 4. On-body real-time message delivery by olfaction system. (A) Schematic demonstration of applying the wireless olfaction interface for the deafblind user. (B) Operational performance of six randomly selected messages delivery of the olfaction system. (C) Average delivery durations of 24 messages by hand-in-hand and olfaction respectively. (D) Recognition rates of 24 messages by hand-in-hand and olfaction respectively. The recognition rates of messages with 24 odorants after 0.25 hr and 24 hr training (E), among 31 volunteers after 0.25 hr and 24 hr training (F), and between genders (G), respectively. Recognition rates of single and the combination of two odorants from eight different OGs (H), among 31 volunteers (I), and between the different training times (0.25 hr and 24 hr) (J), respectively.

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